W.S.72. Friction.

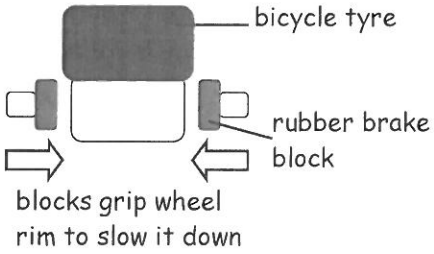
Name .....

Friction is a force that stops two surfaces sliding past each other. It is caused by tiny bumps on the surfaces which catch together.

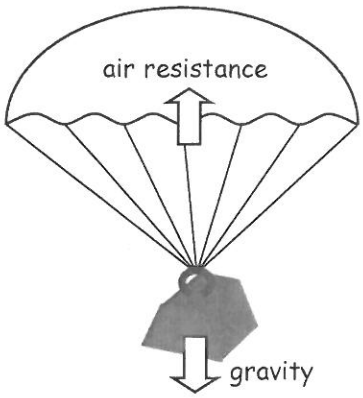
Uses of friction.

1. Friction gives grip for shoes and tyres. We could not move over the ground without friction.

2. Brakes on bicycles and cars use friction to slow down the wheels.



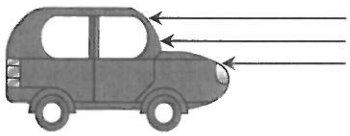
3. Air resistance is a type of friction that slows down parachutes.



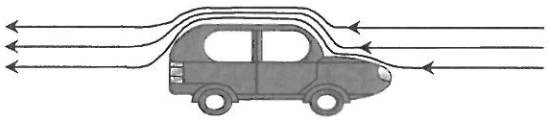
Problems caused by friction.

1. Friction slows down moving machinery. It can also make machinery over heat. Grease and oil must be used to reduce friction.

2. Air resistance is a type of friction that slows down vehicles. The faster the vehicle travels the greater the air resistance becomes. Car bodies are designed so that the air slips smoothly over the bonnet.



Poor design - air hits against bonnet and slows the car down. The engine must work hard to keep at a high speed.



Good design - air slips smoothly over the bonnet. The air resistance is low and the car travels at high speed easily.

Exercise - Complete the missing words in the passage below.

The force that stops two surfaces sliding past each other is called ..... If there was no friction between our shoes and the ground our feet would ..... when we tried to walk. Rubber brake blocks grip against the ..... of a bicycle wheel in order to slow it down. A parachute reaches a steady speed when the force of ..... pulling it down is balanced by the air resistance pushing ..... Friction between the moving parts of machinery can cause it to over ..... The friction can be ..... by using oil or ..... The ..... a car moves the greater the air resistance is that pushes against it.

slip **reduced** **friction** **faster** **heat** **rims** **gravity** **grease** **upwards**

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